AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

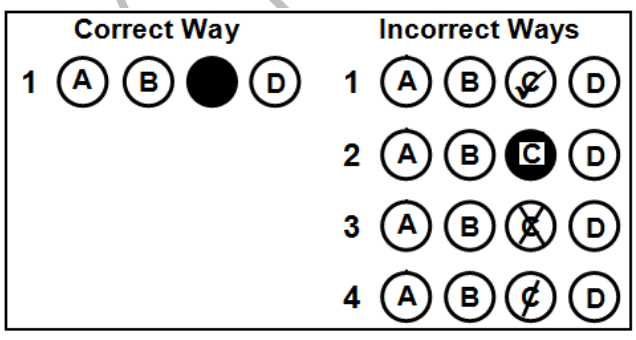
Business Mathematics Paper I

Time: 55 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols have their usual meanings.

Interest and Annuities

$I = \dfrac{PTR}{100}$ $\quad A = P \times \left(1 + \dfrac{r}{n}\right)^{nt}$ $\quad R = \left(1 + \dfrac{i}{n}\right)^{n} - 1$

$j = \left(1 + \dfrac{r}{m}\right)^{m} - 1$ $\quad$ Accumulating factor $= \left(\dfrac{(1+i)^n - 1}{i}\right)$ $\quad$ Annuity $= R \times \left(\dfrac{(1+i)^n - 1}{i}\right)$

Linear Equations, Functions and their Graphs

The general form of linear equation is $ax + by + c = 0$

The intercepts form of the linear equation is $\dfrac{x}{a} + \dfrac{y}{b} = 1$ $a \neq 0$ and $b \neq 0$

Quadratic Equations, Functions and their Graphs

Quadratic Formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Derivative of Algebraic Functions

$\dfrac{d}{dx}[c] = 0$ $\quad \dfrac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \times f'(x)$

$\dfrac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$ $\quad \dfrac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$

$\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

Sequence and Series

$a_n = a_1 + (n-1)d$ $\quad S_n = \dfrac{a_1(1 - r^n)}{1 - r}$, if $|r| < 1$ $\quad S_n = \dfrac{a_1(r^n - 1)}{r - 1}$, if $|r| > 1$

$S_n = \dfrac{n}{2}(2a_1 + (n-1)d)$ $\quad a_n = a_1 r^{n-1}$

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1. A number is increased by 7:5 and it becomes 49. The number was

   A. 37
   B. 35
   C. 20
   D. 12

2. A sum of money is divided between Sardar and Atif in the ratio of 2:3. If Atif gets Rs 6,000, then total amount was

   A. Rs 12,000
   B. Rs 10,000
   C. Rs 9,000
   D. Rs 4,000

3. In the given table, consider the relationship between the values of x and y.

x	3	6	9	12
y	12	6	4	3

   Based on the given information, the variation involved between x and y is

   A. no variation.
   B. direct variation.
   C. inverse variation.
   D. compound variation.

4. If 34% of a number is 69.36, then the number will be

   A. 45.78
   B. 49.01
   C. 204.00
   D. 235.80

5. Shakir bought a bicycle for Rs 30,000 and sold it on 25% profit. The sale price of the bicycle is

   A. Rs 7,500
   B. Rs 22,500
   C. Rs 37,500
   D. Rs 52,500

6. The interest earned on an amount at annual simple interest of 5% in 4 years is Rs 10,000. The principal amount of money is

   A. Rs 2,000
   B. Rs 40,000
   C. Rs 50,000
   D. Rs 200,000

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7. Zara deposits Rs 5,000 in her savings account at the end of each quarter for the next 6 years. Interest is earned at a rate of 3% per year, compounded quarterly.

   The interest rate per quarter is

   A. 0.12
   B. 0.005
   C. 0.0075
   D. 0.00125

8. Imran invested Rs 500,000 at the end of each period for 5 years at 5% interest compounded semi-annually.

   In this case, the number of compounding periods is

   A. 2
   B. 5
   C. 10
   D. 25

9. The slope of the line $2x + 3y = 0$ is

   A. $-\frac{3}{2}$.
   B. $-\frac{2}{3}$.
   C. $\frac{2}{3}$.
   D. $\frac{3}{2}$.

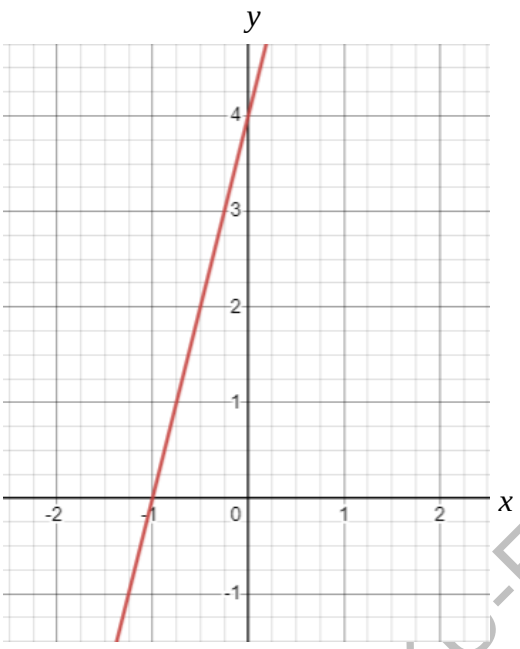
10. The equation of the line parallel to *y*-axis and passing through $(-2, 0)$ will be

    A. $x - 2 = 0$
    B. $y - 2 = 0$
    C. $x + 2 = 0$
    D. $y + 2 = 0$

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Use the given information to answer Q.11 and Q.12.

The equation of a line is represented on the given graph.



11. The slope of the given line is

    A. zero.
    B. positive.
    C. negative.
    D. undefined.

12. The equation of the line represented on the given graph is

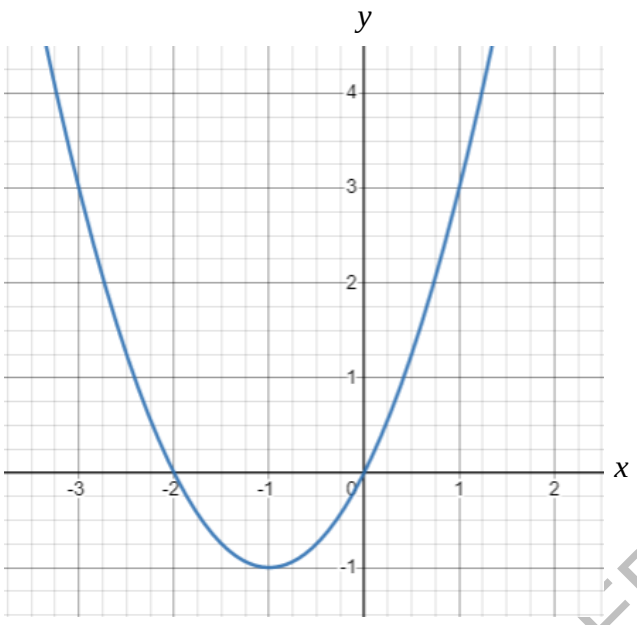
    A. $4x + y + 4 = 0$
    B. $4x - y + 4 = 0$
    C. $4x - y - 4 = 0$
    D. $4x + y - 4 = 0$

13. The solution of the quadratic equation $(x+2)(x-2) = 12$ will be

    A. $\{-2, 2\}$.
    B. $\{-4, 4\}$.
    C. $\{2, 2\}$.
    D. $\{4, 4\}$.

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Use the given graph to answer Q.14 and Q.15.



14. The domain of the quadratic function represented in the given graph is

A. $-2 < x < 0$
B. $-1 < x < 4$
C. $-\infty < x < 0$
D. $-\infty < x < \infty$

15. The minimum value of the quadratic function represented by the given graph is

A. $-\infty$
B. -2
C. -1
D. 0

16. The vertex of the function $y = x^2 - 6x + 10$ lies at the point

A. $(3, -4)$.
B. $(3, 4)$.
C. $(1, 3)$.
D. $(3, 1)$.

17. The solution set of $x + a = 0$ and $y + b = 0$ is

A. (a, b).
B. $\{(a, b)\}$.
C. $(-a, -b)$.
D. $\{(-a, -b)\}$.

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18. The solution set of the simultaneous equations $3x^2 + y^2 = 7$ and $x^2 - 1 = 0$ is

A. $\{(1, 4)\}$.
B. $\{(1, 2)\}$.
C. $\{(\pm 1, \pm 4)\}$.
D. $\{(\pm 1, \pm 2)\}$.

19. On March 20, Zaryab is left with Rs 5,000 from his pocket money. He has to buy 5 T-shirts and 3 pairs of socks for the college tour.

If each T-shirt costs Rs x and each pair of socks costs Rs y, then the mathematical model representing the situation will be

A. $5x + 3y \geq 5000$
B. $5x + 3y \leq 5000$
C. $3x + 5y \geq 5000$
D. $3x + 5y \leq 5000$

20. $\begin{bmatrix} a & c \\ b & -d \end{bmatrix} - \begin{bmatrix} -a & b \\ b & -d \end{bmatrix}$ is equal to

A. $\begin{bmatrix} 2a & 0 \\ 0 & 0 \end{bmatrix}$.
B. $\begin{bmatrix} 2a & c-b \\ 0 & 0 \end{bmatrix}$.
C. $\begin{bmatrix} 0 & c-b \\ 0 & 0 \end{bmatrix}$.
D. $\begin{bmatrix} 0 & c-b \\ 0 & 2d \end{bmatrix}$.

21. The product $E \times F$ for $E = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ and $F = \begin{bmatrix} 0 & 2 \end{bmatrix}$ will be

A. $\begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix}$.
B. $\begin{bmatrix} 0 & 4 \\ 0 & 4 \end{bmatrix}$.
C. $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$.
D. $[4]$.

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22. The determinant of the matrix $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$ will be

A. -1
B. 0
C. 1
D. 2

23. If the determinant of the matrix $\begin{bmatrix} 6 & x \\ 4 & x \end{bmatrix}$ is 8, then the value of x will be

A. -4
B. 0
C. 2
D. 4

24. If $A \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, then the matrix A will be equal to

A. $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$.
B. $\begin{bmatrix} 3 & 3 \\ 1 & 5 \end{bmatrix}$.
C. $\begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}$.
D. $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$.

25. The derivative of $(ax-1)^{\frac{1}{2}}$ with respect to x is

A. $\frac{1}{2}(ax-1)^{\frac{1}{2}}$
B. $\frac{1}{2}(ax-1)^{-\frac{1}{2}}$
C. $\frac{a}{2}(ax-1)^{\frac{1}{2}}$
D. $\frac{a}{2}(ax-1)^{-\frac{1}{2}}$

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26. The derivative of $3\pi^2$ with respect to x will be

A. 0
B. 1
C. 3π
D. 6π

27. If $g(x) = \frac{2}{x-1}$, then $g'(x)$ will be

A. $-\frac{2}{x-1}$.
B. $-\frac{2}{(x-1)^2}$.
C. $\frac{x}{(x-1)}$.
D. $\frac{x}{(x-1)^2}$.

28. The slope of tangent to the curve $y = x^2 - \frac{1}{2}$ at (1, 2) will be

A. 4
B. $\frac{7}{2}$
C. 2
D. $\frac{3}{2}$

29. Which of the given sequences is a geometric sequence?

A. 2, 4, 9, 16, ...
B. 2, 4, 8, 16, ...
C. 2, 4, 6, 10, ...
D. 2, 4, 6, 8, ...

30. If the common difference of an arithmetic sequence is -3 and the 1st term is 5, then the 5th term of the arithmetic sequence would be

A. 8
B. 2
C. -7
D. -10

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END OF PAPER

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